Denoising Diffusion Implicit Models (DDIMs)

Song, Meng & Ermon (ICLR 2021)

Notation

$\boldsymbol{x}_0$: clean data
q : fixed forward process
$\alpha_{1:T} \in (0,1]^T$: decreasing noise schedule
$\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T$: latents
p_θ : parametric (learned) reverse process
$\boldsymbol{\epsilon}_\theta^{(t)}$: noise-prediction network

DDPM Forward Process q (Fixed, Markovian)

$q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) = \prod_{t=1}^T q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})$, where $q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = \mathcal{N}\Big(\sqrt{\frac{\alpha_t}{\alpha_{t-1}}}\,\boldsymbol{x}_{t-1},\ \big(1-\frac{\alpha_t}{\alpha_{t-1}}\big)\boldsymbol{I}\Big)$

Closed-form marginals:

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\big(\sqrt{\alpha_t}\,\boldsymbol{x}_0,\ (1-\alpha_t)\boldsymbol{I}\big) \quad \Leftrightarrow \quad \boldsymbol{x}_t = \sqrt{\alpha_t}\,\boldsymbol{x}_0 + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$$

DDPM Reverse Process p_θ (Learned) and Training

$p_\theta(\boldsymbol{x}_{0:T}) = \mathcal{N}(\boldsymbol{x}_T; \boldsymbol{0}, \boldsymbol{I}) \prod_{t=1}^T p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$, where $p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) = \mathcal{N}\big(\mu_\theta^{(t)}(\boldsymbol{x}_t),\ \sigma_t^2\boldsymbol{I}\big)$.

Training: ELBO gives $L_\gamma(\theta) = \sum_{t=1}^T \gamma_t\, \mathbb{E}_{\boldsymbol{x}_0,\boldsymbol{\epsilon}}\big[\|\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t) - \boldsymbol{\epsilon}\|^2\big]$ L_1**:** the case $\gamma_t = 1\ \forall\, t$

Key observation: L_γ depends only on marginals $q(\boldsymbol{x}_t|\boldsymbol{x}_0)$, *not* on the joint $q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)$.

Non-Markovian Family q_σ

For hyperparameter $\boldsymbol{\sigma} \in \mathbb{R}^T_{\geq 0}$: $q_\sigma(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) = q_\sigma(\boldsymbol{x}_T|\boldsymbol{x}_0) \prod_{t=2}^T q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)$

$$q_\sigma(\boldsymbol{x}_T|\boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_T}\boldsymbol{x}_0, (1-\alpha_T)\boldsymbol{I})$$

$$q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) = \mathcal{N}\left(\sqrt{\alpha_{t-1}}\,\boldsymbol{x}_0 + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\,\frac{\boldsymbol{x}_t - \sqrt{\alpha_t}\,\boldsymbol{x}_0}{\sqrt{1-\alpha_t}},\ \ \sigma_t^2\boldsymbol{I}\right)$$

Lemma: Marginals are preserved: $q_\sigma(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\,\boldsymbol{x}_0, (1-\alpha_t)\boldsymbol{I})$ for all t.

DDIM Generative Process

Denoised estimate: $f_\theta^{(t)}(\boldsymbol{x}_t) := \frac{\boldsymbol{x}_t - \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}{\sqrt{\alpha_t}}$ (predicted $\boldsymbol{x}_0$)

Parametric reverse: $p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t) = q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t,\ f_\theta^{(t)}(\boldsymbol{x}_t)), \quad t > 1$

$$p_\theta^{(1)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_1) = \mathcal{N}(f_\theta^{(1)}(\boldsymbol{x}_1),\ \sigma_1^2\boldsymbol{I})$$

DDIM Update Rule (sampling $\boldsymbol{x}_{t-1}$ from $\boldsymbol{x}_t$, with $\alpha_0 := 1$):

$$\boldsymbol{x}_{t-1} = \sqrt{\alpha_{t-1}} \underbrace{\left(\frac{\boldsymbol{x}_t - \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}{\sqrt{\alpha_t}}\right)}_{\text{“predicted } \boldsymbol{x}_0\text{”}} + \underbrace{\sqrt{1-\alpha_{t-1}-\sigma_t^2}\,\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}_{\text{“direction pointing to } \boldsymbol{x}_t\text{”}} + \underbrace{\sigma_t\boldsymbol{\epsilon}_t}_{\text{noise}}$$

σ_t for various models:

DDPM (η=1):	$\sigma_t = \sqrt{\frac{1-\alpha_{t-1}}{1-\alpha_t}}\sqrt{1-\frac{\alpha_t}{\alpha_{t-1}}}$	stochastic, Markovian
DDIM (η=0):	$\sigma_t = 0\ \ \forall\, t$	deterministic
Interpolation:	$\sigma_t(\eta) = \eta\sqrt{\frac{1-\alpha_{t-1}}{1-\alpha_t}}\sqrt{1-\frac{\alpha_t}{\alpha_{t-1}}}$	$\eta \in (0,1)$

Theorem 1 (Unified Objective)

$$J_\sigma(\theta) := \mathbb{E}_{\boldsymbol{x}_{0:T}\sim q_\sigma}[\log q_\sigma(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) - \log p_\theta(\boldsymbol{x}_{0:T})]$$
$$\equiv \mathbb{E}_{\boldsymbol{x}_{0:T}\sim q_\sigma}\left[\sum_{t=2}^{T} D_{\mathrm{KL}}\big(q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t,\boldsymbol{x}_0)\parallel p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big) - \log p_\theta^{(1)}(\boldsymbol{x}_0|\boldsymbol{x}_1)\right]$$

Theorem 1: $\forall\,\boldsymbol{\sigma} > 0$, $\exists\,\gamma \in \mathbb{R}^T_{>0}$, $C \in \mathbb{R}$ such that $J_\sigma(\theta) = L_\gamma(\theta) + C$.
If optimal θ is independent of $\gamma \Rightarrow L_1$ suffices. **No retraining.**

Accelerated Sampling

Sub-sequence $\tau = [\tau_1, \ldots, \tau_S]$ of $[1, \ldots, T]$ with $S \ll T$. Complement $\bar{\tau} = \{1, \ldots, T\} \setminus \tau$.
Factorized inference:

$$q_{\sigma,\tau}(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) = q_{\sigma,\tau}(\boldsymbol{x}_{\tau_S}|\boldsymbol{x}_0)\prod_{i=1}^{S} q_{\sigma,\tau}(\boldsymbol{x}_{\tau_{i-1}}|\boldsymbol{x}_{\tau_i},\boldsymbol{x}_0)\prod_{t\in\bar{\tau}} q_{\sigma,\tau}(\boldsymbol{x}_t|\boldsymbol{x}_0)$$

$\{\boldsymbol{x}_{\tau_i}\}$ form a **chain** (used for sampling); $\{\boldsymbol{x}_t\}_{t\in\bar{\tau}}$ form a **star** (only in objective).
Generative Process:

$$p_\theta(\boldsymbol{x}_{0:T}) = p_\theta(\boldsymbol{x}_T)\prod_{i=1}^{S} p_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_{i-1}}|\boldsymbol{x}_{\tau_i})\prod_{t\in\bar{\tau}} p_\theta^{(t)}(\boldsymbol{x}_0|\boldsymbol{x}_t)$$

Accelerated update:

$$\boldsymbol{x}_{\tau_{i-1}} = \sqrt{\alpha_{\tau_{i-1}}}\left(\frac{\boldsymbol{x}_{\tau_i} - \sqrt{1-\alpha_{\tau_i}}\,\boldsymbol{\epsilon}_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_i})}{\sqrt{\alpha_{\tau_i}}}\right) + \sqrt{1-\alpha_{\tau_{i-1}}-\sigma_{\tau_i}^2(\eta)}\;\boldsymbol{\epsilon}_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_i}) + \sigma_{\tau_i}(\eta)\,\boldsymbol{\epsilon}$$

where $\sigma_{\tau_i}(\eta) = \eta\sqrt{\frac{1-\alpha_{\tau_{i-1}}}{1-\alpha_{\tau_i}}}\sqrt{1-\frac{\alpha_{\tau_i}}{\alpha_{\tau_{i-1}}}}$.
Subsequence Selection: Linear $\tau_i = \lfloor ci \rfloor$ or Quadratic $\tau_i = \lfloor ci^2 \rfloor$.

Key Properties (DDIM, $\sigma_t = 0$)

- **Deterministic:** $\boldsymbol{x}_0$ depends only on $\boldsymbol{x}_T$. All choices of σ_t use the same pretrained $\boldsymbol{\epsilon}_\theta^{(t)}$.
- **Consistency:** Same $\boldsymbol{x}_T \Rightarrow$ similar high-level features regardless of trajectory length S.
- **Latent interpolation:** Slerp in $\boldsymbol{x}_T$-space gives semantically meaningful image interpolation.

References: Song et al., ICLR 2021 (DDIM)